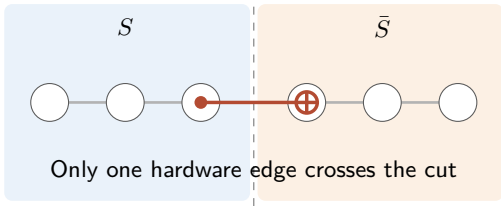


(a) One-dimensional nearest-neighbor chain

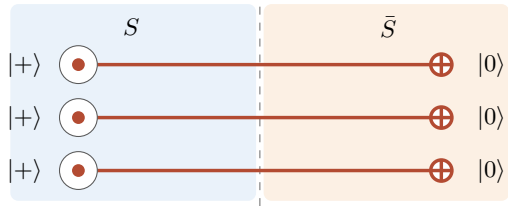


At most one cross-cut CNOT per layer

$$g_S \leq d \implies \text{SR}_{S|\bar{S}} \leq 2^d$$

$d = O(\log n)$: at most $O(\log n)$ entanglement bits

(b) All-to-all connectivity



m disjoint cross-cut CNOTs in one layer

$$d = 1, g_S = m \implies m \text{ Bell pairs}$$

Three pairs shown; m may scale linearly with n